Ramp It Up! 2.0 — Investigation

Chapter 3 · Use with Lesson 4 (Day 1)

Name: _____ Date: _____ Group: _____

BIG QUESTION: How does ramp height affect the change caused by the same ball?

PREDICT

At which ramp height do you think the cup will move the farthest? Why?

FAIR TEST

We change: ramp height (1 book · 2 books · 3 books)

We keep the same: ball · target cup · surface · release point · measurement method

We measure: cup displacement in cm (how far the cup moves)

We do NOT measure speed or energy — we measure the change the ball causes.

DATA TABLE

Run 3 trials at each ramp height. Measure cup displacement in cm. Then find the average.

Ramp Height	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average Cup Displacement (cm)
1 book				
2 books				
3 books				

GRAPH

Ramp Height vs. Average Cup Displacement

Average Cup Displacement (cm)

30			
25			
20			
15			
10			
5			
0			
	1 book	2 books	3 books

Ramp Height (books)

OBSERVE

What pattern do you notice in your data?

INFER

What does the pattern suggest about speed and energy of motion? (Remember: you did not measure speed or energy.)

EXPLAIN

Claim: For the SAME ball, faster motion is associated with greater energy of motion.
Evidence: (What did you measure/observe?)

Reasoning: (How does the evidence support the claim?)
